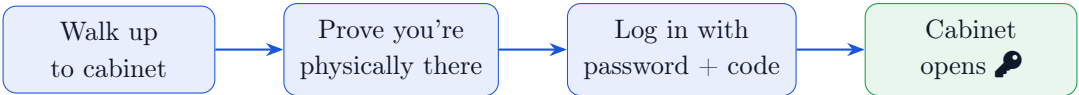


SNTC — Smart Key Storage System

1 What Is Key Management System?

KMS is a **smart cabinet** that manages room keys at IIT Mandi SnTC Clubs. Instead of signing a paper logbook and hoping the key is there, a member walks up to the cabinet, proves they are physically present, logs in with their phone’s authenticator app, and the cabinet opens automatically.



Everything is logged automatically — who took which key, when, and when it was returned. Late returns trigger reminder emails, then escalate to a coordinator.

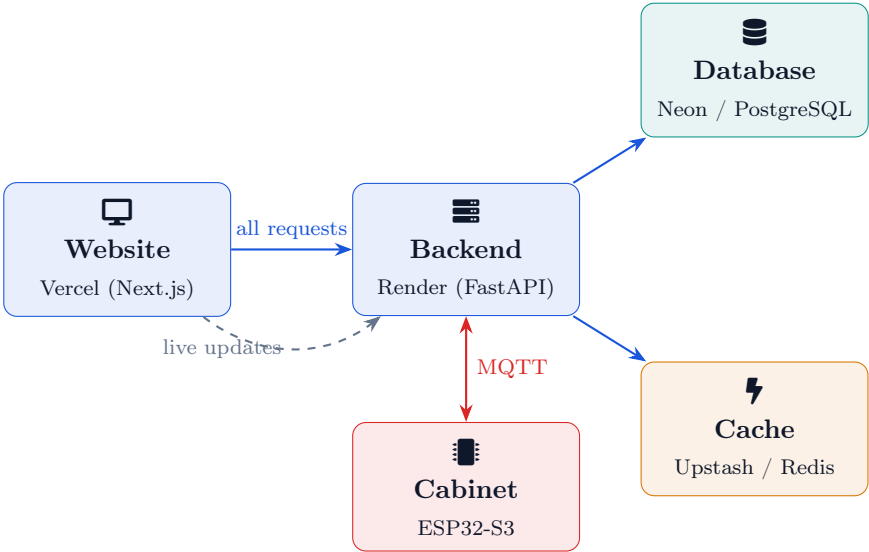
2 Who Uses It?

There are three types of users:

Role	What they can do	Example
Member	Take, return, or extend their own keys	Core Member
Coordinator	All of the above + see logs for their rooms, override issues	Co-ordinator
Admin	Everything — manage users, devices, generate reports	Dean

3 How the System Is Built

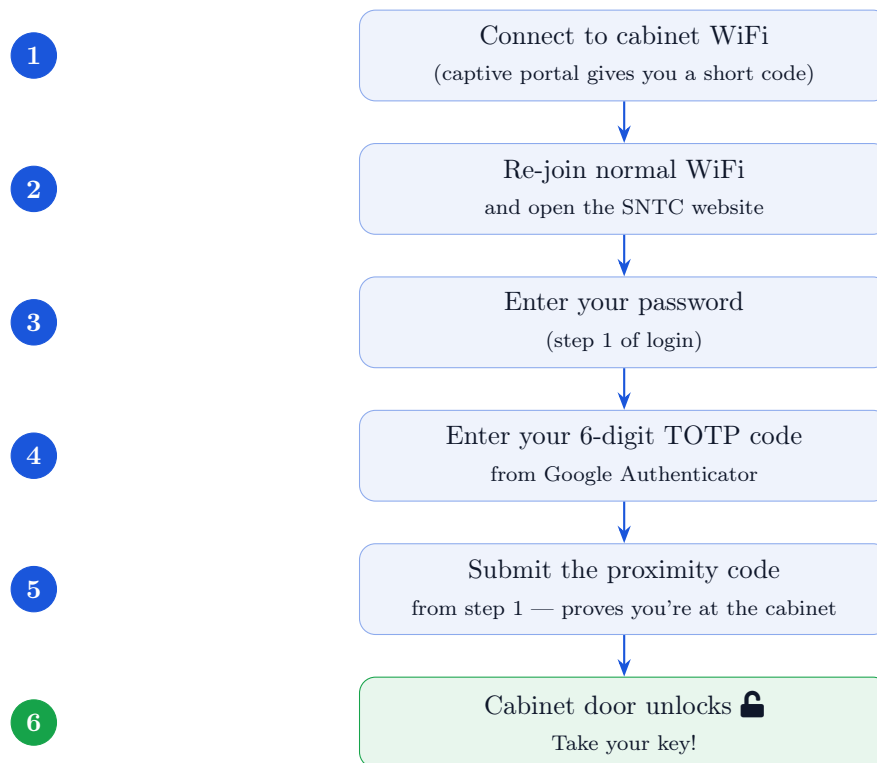
KMS has four main pieces that work together:



- **Website** — what members and admins see in their browser. Hosted on Vercel, built with Next.js.
- **Backend** — the brains. Checks passwords, controls the cabinet, sends emails. Hosted on Render, built with FastAPI (Python).
- **Database + Cache** — stores all records permanently (PostgreSQL) and keeps fast temporary data like login sessions (Redis).
- **Cabinet hardware** — an ESP32-S3 microcontroller with a servo motor that physically locks and unlocks key slots.

4 Logging In — Step by Step

Security is taken seriously here: you need **three things** to get a key — something you know (password), something you have (your phone with the authenticator app), and physical presence near the cabinet.



i Note

The 6-digit TOTP code changes every 30 seconds. Use Google Authenticator or any standard authenticator app. You only need to scan the QR code once when you first set up your account.

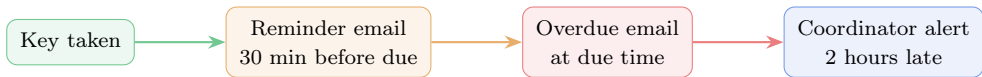
5 Key Actions

Once logged in and near the cabinet, members can do three things:



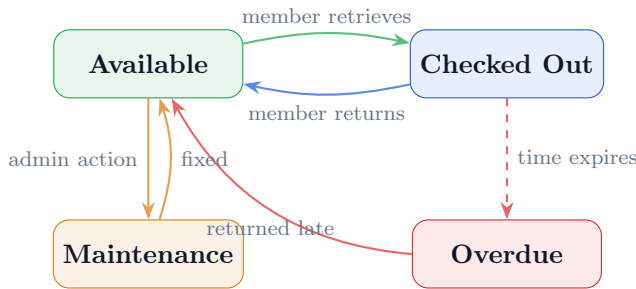
Every key has a default 6-hour possession window. The system sends an email reminder 30 minutes before that window closes.

5.1 What happens if a key is kept too long?



6 Key States

Each key slot is always in one of four states:



7 Security at a Glance

What’s protected	How
Passwords	Stored as a one-way hash (bcrypt) — even admins cannot read them
Login	Requires both your password <i>and</i> a 30-second phone code
Too many tries	Locked out after 5 wrong attempts
Sessions	Expire automatically after 1 hour of inactivity
Key retrieval	Must be physically near the cabinet (proximity flag, 5 min window)
Cabinet unlock	Each command has a one-time code — replaying it does nothing
Audit trail	Every action is permanently logged; nothing can be deleted

8 Website Pages

Page	Who uses it	What it does
/login	Everyone	Enter password then TOTP code
/connect	Everyone	Submit the proximity code from the cabinet WiFi
/keys	Members	See all keys, take / return / extend
/admin	Coordinators & Admins	Manage users, view logs, run reports

The key grid on `/keys` updates **live** — if someone else returns a key while you are looking at the page, it shows as available immediately, without refreshing.

9 API Quick Reference

These are the main server endpoints. Each badge shows the HTTP method.

Endpoint	What it does
POST /auth/login	Check password, return a temporary token
POST /auth/totp/verify	Check the 6-digit code, set login session
POST /auth/logout	End the session
POST /proximity/verify	Submit the cabinet WiFi code
POST /sessions/start	Unlock the cabinet door
GET /keys	List all key slots
POST /keys/{slot}/retrieve	Take a key
POST /keys/{slot}/return	Return a key
POST /keys/{slot}/extend	Ask for more time
GET /admin/dashboard	Admin overview — overdue keys, recent events
GET /admin/logs/retrieval	Full key-retrieval history
POST /admin/logs/override	Manually override a key state
GET /health	Check the server is running
WS /ws/keys	Live key-grid updates

10 Deployment Overview

The system will be hosted across four cloud services. Set them up in order:



⚠ Watch out

Render's free plan pauses after 15 minutes of no traffic. This stops scheduled emails. Will setup free uptime monitor (e.g. Cronjob, Uptimebot) at `/health` to keep it awake.

Summary

✓ Summary

KMS in one paragraph:

KMS is a self-service key cabinet for IIT Mandi SnTC Clubs that replaces paper sign-out sheets with a secure, fully automated system. A member connects to the cabinet's WiFi to get a one-time proximity code, then logs in on the website with their password and a phone-based one-time code. Only after all three checks pass does the cabinet open. The system records every action, sends email reminders before keys are due, and escalates overdue keys to a coordinator automatically. Admins can manage users, rooms, and devices through a web dashboard. The whole stack runs on Vercel (frontend), Render (backend), Neon (database), and Upstash (cache) — and can also be self-hosted with Docker Compose.

Feature	Detail
Authentication	Password + TOTP (2-factor)
Proximity check	Must be within WiFi range of cabinet
Key window	6 hours default; extendable
Reminder email	30 min before due
Overdue escalation	Coordinator notified 2 hours after due time
Audit log	Permanent, tamper-evident
Live key grid	Updates via WebSocket — no page refresh needed
Roles	Member, Coordinator, Admin
Self-hosted option	<code>docker compose up</code>